

Pulse — from BOSS history to a bid recommendation

The concierge methodology, documented from its first two live cases - August 2026

What this is. Pulse is a bid-optimization tool for SMU's BOSS course bidding. Before writing the software, we're running it as a *concierge MVP*: the full analysis is performed by hand for real students with real e-dollars on the line, so that the product is designed around what the analysis actually requires - not what we guessed it would. This document records that process end to end.

Step 1 · Collect the raw history

Source: the Overall BOSS Results table, captured per course across at least two academic years. The rounds that matter are the ones the student will actually bid in - for a freshman, the Incoming Freshmen Rnd 1 windows; senior rounds (1A/1B/2) use different quota pools and player types and mostly add noise. Per row we need: Term, Bidding Window, Section, Median and Min successful bid, Vacancy, Open, Bef Proc, Aft Proc, Enrolled, Instructor. Screenshots are fine; there is no export button. Our two cases so far ran on 303 rows (COR1703) and 1,102 rows (STAT1202).

Step 2 · Decode the schema - and verify it

BOSS doesn't document its columns, so we decode them and then try to break the decoding. Working interpretation: **Bef Proc / Aft Proc are vacancies before and after the window is processed**, so cleared bids = Bef – Aft and Enrolled = quota – Aft Proc. Verification is by falsification: under the rival interpretation (bids placed / bids failed), one STAT1202 section would have had 34 bids "fail" into 48 empty seats - impossible - while our interpretation reproduces the Enrolled column exactly across every freshman row in both datasets. Two practical corollaries: the class-search "Available" number is a ceiling, not a promise (verify seat release against what actually cleared), and **if Aft Proc > 0, every bid in that section cleared** - which lets us detect windows with zero failures.

Step 3 · Reconstruct the current state

From the newest window's rows we build the board: seats remaining per section, bids cleared, clearing prices, and - critically - where failures happened. Failed bidders are the most predictable demand in the next window: they've revealed both their intent and their willingness to pay. A course where nobody failed (COR1703: every section had leftovers) faces only fresh entrants next window; a course with eleven full sections clearing at e\$27-42 (STAT1202) has a rebidder wave incoming.

Step 4 · Join history on the right key

Section letters are not stable across years - last year's "G9" can be a different professor, timeslot, and demand profile entirely. History must be joined on **instructor** (and where known, timeslot). This also surfaces genuine blind spots honestly: a professor new to the course has no history, and their sections must be treated as wildcards rather than silently inheriting a stranger's price record.

Step 5 · Estimate next-window demand

Three ingredients: (1) a **carryover ratio** - how many bidders historically appear in Window N+1 per Window-N bidder (≈ 0.25 -0.33 in our data), scaled to this year's observed Window-1 volume; (2) the **rebidder wave** from Step 3; (3) an allocation model spreading those bidders across sections by a blend of instructor popularity (revealed by cleared bids and medians) and seat availability (demand flows toward supply). Every ingredient is uncertain, so each becomes a scenario axis rather than a point estimate: cold / base / hot demand, and - the one that matters most - **targeted rush** scenarios where locked-out bidders converge on a single attractive section instead of spreading politely.

Step 6 · Monte Carlo simulation

We simulate the window 20,000-40,000 times. Each run: draw a rival count per section (Poisson around the allocation estimate), draw each rival's bid (lognormal above the e\$10 floor, anchored on that section's last-window median - people anchor on the last price they saw), then apply BOSS's rules: highest bids take the seats. The mechanic to internalise is **musical chairs, not price tags**: if rivals < seats, every bid wins and the amount is irrelevant; only when a section overfills does a *clearing price* exist - the bid of the last person seated. Outputs per section: the probability any candidate bid clears, P(section fills at all), and the clearing-price distribution summarised as p50/p90/p99 - bid above p99 and you survive all but the weirdest 1% of simulated futures.

Step 7 · Stress-test instead of trust

A simulation can be confidently wrong: 40,000 runs of bad assumptions agree with each other beautifully. So no recommendation ships on the base model alone. We re-run with the assumptions deliberately broken against us - price anchors inflated by the historically observed window-over-window markup (~1.3-1.45x), fatter tails so real-world freak bids occur regularly, demand cranked past anything observed, and rushes aimed directly at the recommended section. A recommendation is only made when it survives the assumptions being wrong, and the one number that finally sets the bid is usually a **real precedent**: e.g. "price above the worst thing any comparable section has ever done" (STAT1202's G8 rec of e\$30-32 sits above the e\$25.88 minimum at which last year's biggest section was stamped full).

Step 8 · Decision logic, not just probabilities

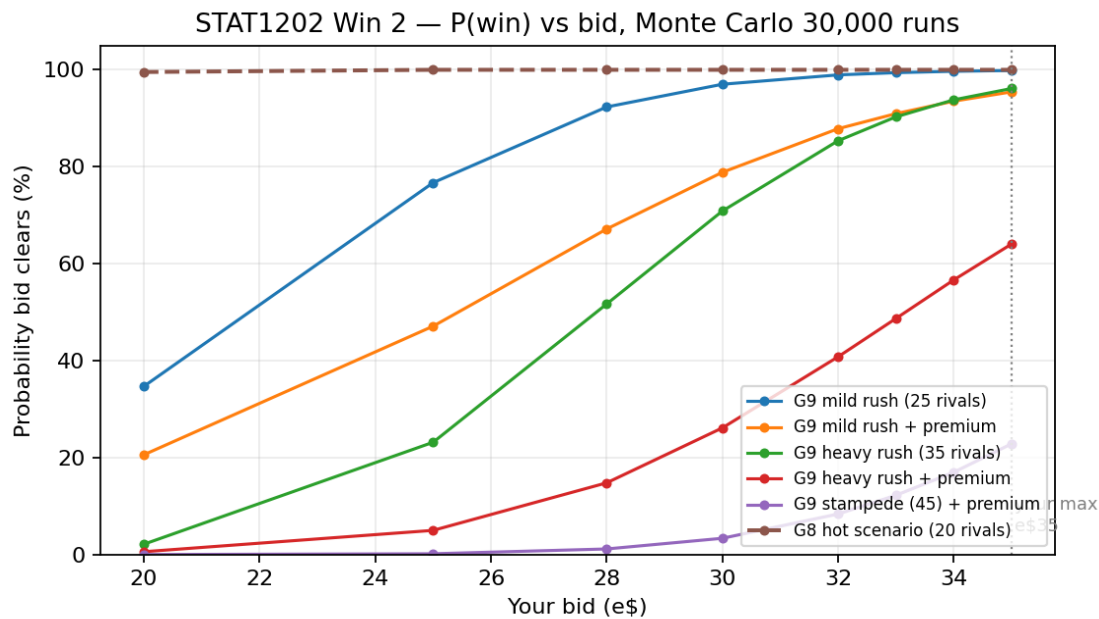
Probabilities become advice through a few rules. **Constraint triangle**: when preferences (professor, timeslot, certainty) can't all fit the budget, we make the trade-off explicit and let the student pick two. **Bid sizing**: failed bids refund fully, so a must-have single bid goes all-in, while a near-certain bid is shaded to the stress-tested number since unspent e\$ rolls forward to future, possibly harder, auctions. **Never floor-bid a tight section** - the only reliably losable move. **Dominance check**: the strongest recommendations are undominated - in the scenarios where they fail, no alternative use of the budget would have won either. **Calibration from failure**: a student's own failed bid (amount vs the published Min Succ Bid) is the best single data point for how they misjudge contested sections.

Step 9 · Deliver probabilities, not promises

The output is a short report: recommendation first, reasoning after, uncertainty always. "100%" is always glossed as "never failed in N simulated futures sharing our assumptions" - never as a guarantee. Each report ends with a morning-of checklist (live seat counts, timeslot confirmation, exact balance), a named fallback (later windows' historical prices), and the failure modes spelled out. This framing is deliberate: it keeps the analyst an analyst rather than a guarantor, and it is the disclaimer architecture the eventual product will ship with.

The two live cases (contrast is the point)

	Case 1 - COR1703 (VUCA)	Case 2 - STAT1202 (Intro Stats)
Regime	Abundant: 106 seats left, 66 Win-1 bidders, zero failures	Contested: ~680 Win-1 bids cleared, 11/25 sections full, prices to e\$42
Key insight	No failures → no rebidder wave; bid amount nearly irrelevant	Refugee wave from full sections; budget sits inside historical clearing range
Stress result	Robust: e\$15 survived 1.45x premium + fat tails everywhere but one 6-seat section	Split: target section 23-100% depending on rush; sibling section survives all rushes
Recommendation	e\$15 any section (e\$20 for the 6-seat one)	e\$30-32 on the Monday sibling, or all-in e\$35 on Wednesday accepting rush risk



Example output artifact (Case 2): win probability vs bid under escalating rush scenarios - the shape of the answer a Pulse user sees.

Known limitations (kept on purpose)

(1) **Regime changes:** the model extrapolates behaviour and cannot see news - timetable shuffles, quota changes, concurrent exchange-round seat consumption. (2) **Clumpy demand:** friends bid in groups; Poisson arrivals understate the chance of a coordinated pile-on. (3) **Circularity:** stress tests prove robustness within the model family, not truth - the claims that stand on raw data alone (seats, cleared counts, historical extremes) carry the real weight. (4) **Reflexivity:** if many bidders run this same analysis, cheap sections stop being cheap - the predictions can self-defeat at scale, which is a product-design problem Pulse will eventually have to price in.

Pulse · concierge preview · analysis on historical BOSS data, delivered as probabilities with named failure modes. Not affiliated with SMU.